Free particle propagator 1

Let $K(b,a)$ be the amplitude for a particle to go from a to b where $\hat{H}$ is the Hamiltonian.

$$K(b,a) = \langle x_b | \exp\left(-\frac{i}{\hbar}(t_b - t_a)\hat{H}\right) |x_a\rangle$$

For a free particle

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

Hence

$$K(b,a) = \langle x_b | \exp\left(-\frac{i}{\hbar}\frac{(t_b - t_a)}{2m}\hat{p}^2\right) |x_a\rangle$$

By the identity

$$\int_{-\infty}^{\infty} |p\rangle\langle p| \, dp = 1$$

we can write

$$K(b,a) = \int_{-\infty}^{\infty} \langle x_b | \exp\left(-\frac{i}{\hbar}\frac{(t_b - t_a)}{2m}\hat{p}^2\right) |p\rangle\langle p|x_a\rangle \, dp$$

By the formula $\hat{p}|p\rangle = p|p\rangle$ we can substitute eigenvalue p for momentum operator $\hat{p}$.

$$K(b,a) = \int \langle x_b | \exp\left(-\frac{i}{\hbar}\frac{(t_b - t_a)}{2m}p^2\right) |p\rangle\langle p|x_a\rangle \, dp$$

Now that the operator $\hat{p}$ is gone we can rearrange as

$$K(b,a) = \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar}\frac{(t_b - t_a)}{2m}p^2\right) \langle x_b|p\rangle\langle p|x_a\rangle \, dp$$

For a free particle

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{ipx}{\hbar}\right)$$

Hence

$$K(b,a) = \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar}\frac{(t_b - t_a)}{2m}p^2\right) \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{ipx_b}{\hbar}\right) \frac{1}{\sqrt{2\pi\hbar}} \exp\left(-\frac{ipx_a}{\hbar}\right) \, dp$$

Simplify as

$$K(b,a) = \frac{1}{2m\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar}\frac{(t_b - t_a)}{2m}p^2 + \frac{i}{\hbar}(x_b - x_a)p\right) \, dp$$

Let

$$a = \frac{i}{\hbar}\frac{(t_b - t_a)}{2m}$$

and

$$b = \frac{i}{\hbar}(x_b - x_a)$$

so that

$$K(b,a) = \frac{1}{2m\hbar}\int_{-\infty}^{\infty} \exp\left(-ap^2 + bp\right)\, dp$$

By the Gaussian integral formula

$$\int_{-\infty}^{\infty} \exp\left(-ap^2 + bp\right)\, dp = \sqrt{\frac{\pi}{a}}\exp\left(\frac{b^2}{4a}\right)$$

we have

$$K(b,a) = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m\hbar}{i(t_b - t_a)}}\exp\left(-\frac{m(x_b - x_a)^2}{2i\hbar(t_b - t_a)}\right)$$

Rearrange as

$$K(b,a) = \sqrt{\frac{m}{2\pi i\hbar(t_b - t_a)}}\exp\left(\frac{im(x_b - x_a)^2}{2\hbar(t_b - t_a)}\right) \tag{1}$$